Experiment No-16 : Feed Forward Neural Network

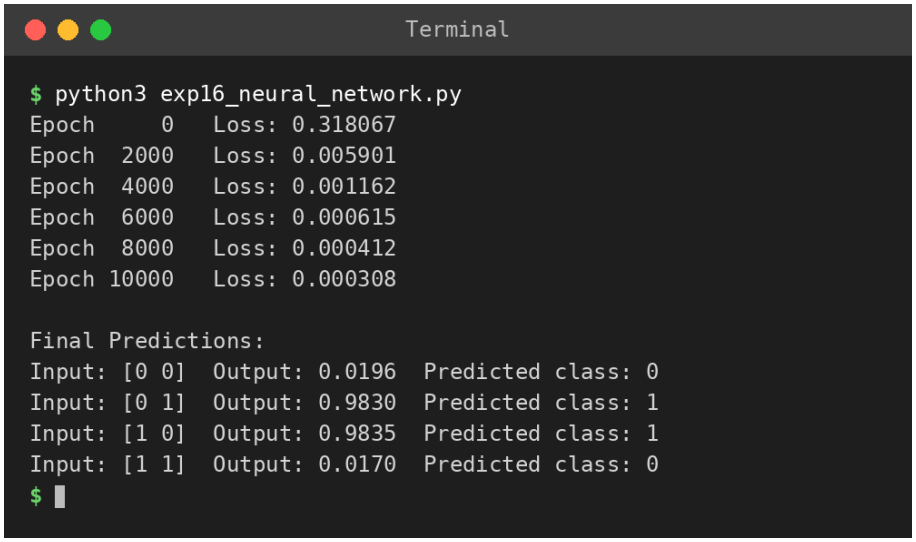
Aim

To write a Python program to implement a Feed Forward Neural Network with back-propagation in Artificial Intelligence, and to train it to solve the XOR problem.

Programming Code:

```
import numpy as np

np.random.seed(1)

# XOR dataset
X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])
y = np.array([[0], [1], [1], [0]])

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def sigmoid_derivative(a):
    return a * (1 - a)

# Network: 2 input -> 4 hidden -> 1 output
W1 = np.random.uniform(size=(2, 4))
b1 = np.zeros((1, 4))
W2 = np.random.uniform(size=(4, 1))
b2 = np.zeros((1, 1))
lr = 0.5

for epoch in range(10001):
    # Forward propagation
    hidden = sigmoid(np.dot(X, W1) + b1)
    output = sigmoid(np.dot(hidden, W2) + b2)

    # Error
    error = y - output
    loss = np.mean(error ** 2)

    # Backward propagation
    d_output = error * sigmoid_derivative(output)
    d_hidden = np.dot(d_output, W2.T) * sigmoid_derivative(hidden)

    # Update weights and biases
    W2 += np.dot(hidden.T, d_output) * lr
    b2 += np.sum(d_output, axis=0, keepdims=True) * lr
    W1 += np.dot(X.T, d_hidden) * lr
    b1 += np.sum(d_hidden, axis=0, keepdims=True) * lr

    if epoch % 2000 == 0:
        print(f"Epoch {epoch:5d}   Loss: {loss:.6f}")

print("\nFinal Predictions:")
for inp, out in zip(X, output):
    print(f"Input: {inp}  Output: {out[0]:.4f}  Predicted class: {int(out[0] > 0.5)}")
```

OUTPUT:



Expected Output

- The loss (mean squared error) is printed every 2000 epochs. It starts at 0.318067 in epoch 0 and falls to 0.000308 at epoch 10000, which shows that the network is learning.
- The final predictions are listed for all four inputs of the XOR truth table.
- Inputs [0 0] and [1 1] give outputs close to 0 (0.0196 and 0.0170), so the predicted class is 0.
- Inputs [0 1] and [1 0] give outputs close to 1 (0.9830 and 0.9835), so the predicted class is 1.
- The exact decimal values may differ slightly with a different random seed, but the predicted classes will be 0, 1, 1, 0.

Result

Thus, the Feed Forward Neural Network was implemented successfully in Python. The network was trained using back-propagation and learned the XOR function correctly.